CHAPTER 19

INTERCALATION AND RELEASE OF DINUCLEOTIDES FROM A NANODIMENSIONAL LAYERED DOUBLE HYDROXIDE

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ABSTRACT

Diadenosine-5′,5‴-P^1, P^4-tetraphosphate (Ap_4A) and bis-(3′-5′)-cyclic dimeric deoxyguanosine monophosphate (c-di-GMP) intercalated into nanosized (200–300 nm) Al, Mg layered double hydroxides (LDHs) were originally prepared. Binding of both dinucleotides with inorganic carrier ranged from 0.8 to 1 μmol/mg LDH. Immobilization process was shown not to alter chemical structure of the dinucleotides. Parameters of Ap_4A and c-di-GMP elution from nanoparticles were studied. It was found that rate of dinucleotide release from LDH nanocomplexes (LDH/Ap_4A and LDH/c-di-GMP) varied depending on pH of the media (higher at acid pH and lower at neutral pH values). The obtained results evidence in favor of using LDH/Ap_4A and LDH/c-di-GMP nanocomposites as pH-controlled passive system for delivery of pharmaceutically valuable dinucleotides into target cells.

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19.1 INTRODUCTION

Delivery of pharmaceuticals to affected tissues envisages handling problems related to preservation of their structure and activity along the transit route from introduction zone to the target cells and further translocation through cell membrane. Moreover, the drug supplied by conventional means tends to be evenly distributed in the body, reaching not only the target cells where it is expected to display therapeutic effect but also other vital systems causing potentially adverse impact. It appears natural therefore that currently the keen interest of researchers has been focused on elaboration of methods for passive or precise transportation of medicines into cells using organic and inorganic nanovehicles [1–3].

Application of nanoparticles as drug carriers enables to reduce considerably systemic intoxication of the organism as compared to molecular form, to protect active substances from premature decomposition in blood stream, to modify their pharmacokinetic parameters and to promote their penetration through plasmatic membrane and intracellular partitioning [4].

Vivid interest around the world has been aroused by systems of drug delivery engaging nanodimensional inorganic supports [5]. It was provoked by discovery of special wall characteristic in vessels feeding tissues exposed to cancerogenesis, inflammation, infarction, hypoxia, etc.

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denoted as hyper transparent capacity, and with respect to tumor cells – enhanced permeability and retention of inflowing substances (termed EPR effect) [6, 7]. Unlike normal, such blood vessels are distinguished by loose structure passing relatively large molecules (about 500 nm). The described phenomenon lays the ground for engineering drug-nano-carrier composites as constituents of passive delivery system.

A whole spectrum of materials was proposed for construction of such drug-courier nanocomplexes – calcium phosphate, gold, carbonaceous compounds, silica, ferric oxide, and layered double hydroxides (LDHs) [8, 9].

LDHs, also called "anionic clays" represent a family of materials that have attracted increasing attention in recent years due to their technological importance in catalysis, separation technology, optics, medical science, and nanocomposite engineering. LDHs consist of positively charged metal hydroxide layers and interlayer gallery, in which the anions (along with water) are stabilized in order to compensate the positive layer charges. The general chemical formula of LDH clays is expressed as $[M^{II}_{1-x}M^{III}_{x}(OH)_2]^{x+}(A^{n-})_{x/n}\cdot yH_2O$, where M^{II} is a divalent metal ion, such as Mg^{2+}, Ca^{2+}, Zn^{2+}, etc., M^{III} is a trivalent metal ion, such as Al^{3+}, Cr^{3+}, Fe^{3+}, Co^{3+}, etc. and A^{n-} is an anion of any type with charge n, such as Cl^-, CO_3^{2-}, NO_3^-, etc. [10]. The electrostatic interactions, hydrogen bonds between layers and contents of the gallery hold the layers together, making a three-dimensional structure (Figure 19.1). There are several combinations of divalent and trivalent cations that can form LDHs. For these ions, the only requirement is that their radii are not too different from those of Mg^{2+} and Al^{3+}. The anions occupy the interlayer gallery region

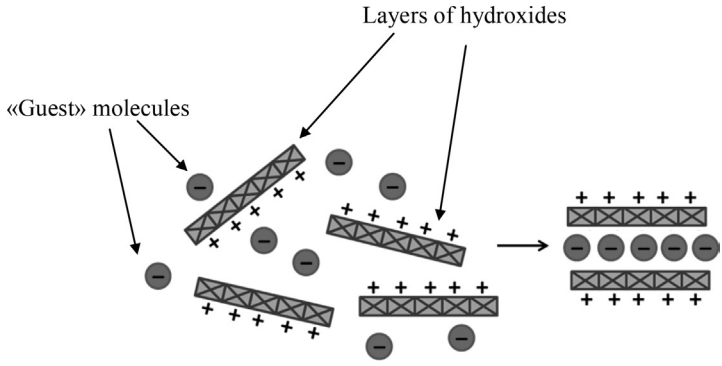


FIGURE 19.1 Schematic structure of layered double hydroxides.

of these crystalline materials. The remarkable behavior of LDHs is their high reactivity toward various organic anions, which can exchange as much as 80–100% of the interlayer molecules.

LDHs have historically been of interest as catalysts, ceramic precursors, traps for anionic pollutants, catalyst supports, ion exchangers, and additives for polymers. More recently, the successful synthesis of LDH materials on the nano scale has paved the way for many novel applications particularly in the field of nanomedicine. For example, LDH materials are increasingly explored as controlled release systems [8, 11, 12]. Up to now many kinds of anions were intercalated in the LDH interlayer gallery, such as common inorganic anions, organic anions, polymeric anions, complex anions, macrocyclic ligands and their metal complexes, polyoxalates, and biochemical anions (amino acids, CMP, AMP, ATP and nucleic acid fragments) [13, 14].

LDHs possess a series of advantages as transporters over other nanocarriers. They display several beneficial properties, including primarily ready bioavailability and low toxicity. According to literature reports, synthesis of LDHs differing in qualitative and quantitative composition of metal cations yields under definite conditions nanoparticles ranging in size from 50 to 500 nm [15]. The substance embedded in interlayer spacing is safely protected from deteriorating impact of damaging factors [16].

LDHs carry a minor positive charge facilitating efficient intercalation of anions and sufficient electrostatic attraction to negatively charged cell surface [8]. In addition, LDH particles rapidly decay in media with low pH values to relatively non-toxic kidney-evacuated metal ions (e.g., Mg^{2+} and Al^{3+}). It is vital that LDHs retain stability at neutral pH values.

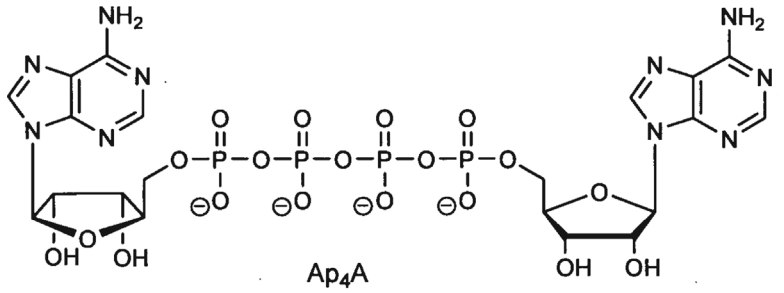
In the present study, we prepared artificially novel bio-inorganic nanohybrids of LDH and pharmaceutically valuable biomolecules Ap_4A and c-di-GMP (Figure 19.2) and investigated the possibility of using intercalated LDH as a drug delivery system.

19.2 MATERIALS AND METHODS

19.2.1 MATERIALS

Ap_4A sample was synthesized from ATP enzymatically as previously reported [17]. Synthesis of c-di-GMP was described in our paper earlier [18].

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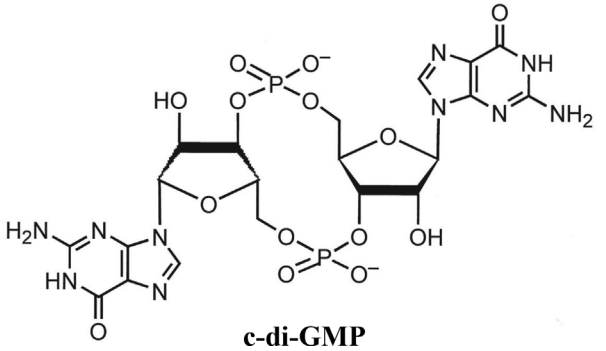


FIGURE 19.2 Structure of the dinucleotides, Ap_4A and c-di-GMP.

ATP, AMP, adenosine and adenine were purchased from Sigma-Aldrich (USA). Twice-distilled water from which carbon dioxide was removed by boiling was used in all experiments.

19.2.2 PREPARATION OF LDH/AP$_4$A AND LDH/C-DI-GMP NANOCOMPOSITE PARTICLES

The nanocomposites with Mg:Al ratio 2:1 were synthesized by coprecipitation method according to [19], with subsequent crystallization at 75°C in the presence of ammonium hydroxide. Briefly, 10 mL salt solution containing 0.02 M $Mg(NO_3)_2$ and 0.01 M $Al(NO_3)_3$ was added drop-wise with vigorous stirring at 25°C to 10 mL of 10 mM NH_4OH solution comprising 10 mM Ap_4A or c-di-GMP. The reaction slurry was aged at 75°C for 12 h.

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The resulting white precipitate was collected by centrifugation (20 000 g for 10 min), washed five times with water, and finally, with acetone. All the samples were air-dried at 60°C for 5–6 h.

19.2.3 CHARACTERIZATION OF LDH/AP$_4$A NANOHYBRIDS

The obtained LDH nanoparticles were imaged using a JSM-2010 (JEOL, Japan) transmission electron microscope (TEM) at the acceleration voltage 200 kV. The loading capacity of LDH/Ap$_4$A and LDH/c-di-GMP nanoparticles and the efficiency of this process under different experimental conditions were evaluated using the following method. A known amount of the nanocomposites was placed in a 10 mL volumetric flask and LDH layers were dissolved in 0.02 M HCl solution. Concentration of Ap$_4$A or c-di-GMP in the solution was determined by monitoring the absorbance at λ=260 nm for Ap$_4$A (ε=30800 $M^{-1}\cdot cm^{-1}$) or λ=253 nm for c-di-GMP (ε=23700 $M^{-1}\cdot cm^{-1}$) with Shimadzu Corporation model UV-1202 spectrophotometer.

19.2.4 RELEASE OF DINUCLEOTIDES FROM LDH/AP$_4$A OR LDH/C-DI-GMP NANOHYBRIDS

To measure the amount of Ap$_4$A and c-di-GMP released from the nanohybrids, the in vitro drug release tests were performed at 25°C by stirring powdered LDH/Ap$_4$A or LDH/c-di-GMP nanohybrids (0.05 g) in 20 mL of 0.05 M phosphate-citrate buffer solution (pH 4.4 or 7.5, respectively) [20]. Aliquots (1 ml) of the suspension were sampled at desired time intervals, centrifuged and the Ap$_4$A and c-di-GMP contents of supernatant were determined by reading UV absorbance at λ=260 nm for Ap$_4$A or λ=253 nm for c-di-GMP to calculate the amount of dinucleotides released from the nanohybrids. The percentage released at each time point was expressed as a fraction of the total amount of the dinucleotides.

19.3 RESULTS AND DISCUSSION

LDHs are currently produced, as a rule, by one of the following methods: (i) coprecipitation [21]; (ii) ion exchange [22]. Co-precipitation regarded

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as preferential technique envisages adding alkaline solution to the mixture containing divalent and trivalent metal cations. As to particulate dimensions, it should be stated that the size and morphology of LDH nanoparticles depend on conditions of synthesis and nature of gallery (guest) molecules, while precise correlation of morphological changes with LDH composition remains to be established [23].

The LDHs, containing Ap_4A (LDH/Ap_4A) or c-di-GMP (LDH/c-di-GMP) were prepared by coprecipitation from a mixed aqueous solution containing Mg^{2+}, Al^{3+} and the dinucleotides. In order to choose the optimal conditions for intercalation, various reaction parameters, like temperature, time, concentration of the components and pH, were examined. Based on these experiments, it was concluded that the optimal intercalation conditions for Ap_4A are as follows: Mg:Al molar ratio 2:1, room temperature, pH 10.0, and 12 h aging. The maximum amount of Ap_4A in the intercalated compound was around 0.83 mmol per 1 g of LDH-nanocomposite.

The similar c-di-GMP immobilization procedure allowed to achieve drug loading approximately 1 mmol per 1 g of LDH-nanohybrid (Table 19.1). For comparison, the Table 19.1 provides data on immobilization of additional nucleic acid compounds – ATP, AMP, adenosine and adenine.

It is clear from Table 19.1 that the number of guest molecules intercalated into LDHs is a function of several factors. The key parameter is molecular charge. For instance, adenine and adenosine lacking negative charge are not intercalated into LDH structure. On the other hand, elevated negative charge results in reduced number of attracted molecules, probably due to deficiency of available positive charges in LDHs [24]. It is well illustrated by comparative amounts of intercalated AMP and ATP (Ap_4A) – 1.44 versus

TABLE 19.1 The Loading of the Guest Molecules in LDH-Nanocomposites

Guest molecule	Guest (mmol/g)	Guest (g/g)	Loading percentage (w/w)
C-di-GMP	1.1±0,05	0.81±0,04	81
Ap_4A	0.83±0,03	0.69±0,03	69
ATP	0.79±0,03	0.38±0,02	38
AMP	1.44±0,07	0.50±0,02	50
Adenosine	–	–	–
Adenine	–	–	–

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0.79 (0.83) mmol/g LDH. The other factors governing incorporation of guest compounds into LDH structure are the form and size of the molecule, its hydrophobicity, solubility of LDH/guest complex [24].

The morphology and dimensions of LDH/Ap_4A nanohybrids have been estimated by electron microscopy. TEM image of LDH/Ap_4A nanoparticles is presented in Figure 19.3. It should be noted that presence of small hexagonal platelets having a diameter of about 200 nm together with some larger platelets (about 350 nm), could be explained by formation of aggregates. Microphotographs of LDH/c-di-GMP nanoparticles resembled images of LDH/Ap_4A (data not provided).

The intercalated dinucleotides were quantitatively recovered from the host lattices by treating LDH/Ap_4A and LDH/c-di-GMP nanoparticles with 0.02 M HCl [24]. This technique coupled to high performance liquid chromatography (LKB, Sweden; column Ultropac TSK ODS-120T) of guest molecules before and after elution from LDH complex (Figure 19.4) led us to assume that Ap_4A and c-di-GMP compounds were intercalated into LDH in intact form.

In order to investigate the possibility of using intercalated LDH nanohybrid as a drug delivery system, deintercalation of Ap_4A and c-di-GMP was examined. Typical kinetic curves of Ap_4A and c-di-GMP release from the nanocomposites at different pH values are shown in Figure 19.5. Noteworthy,

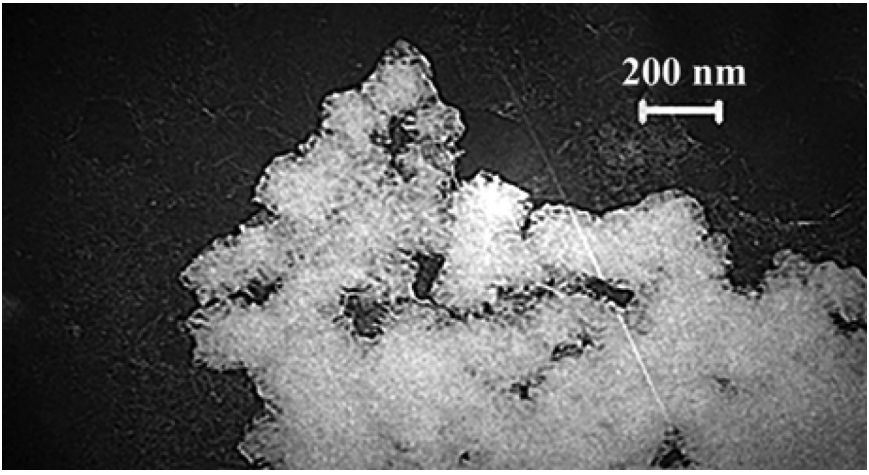


FIGURE 19.3 Transmission electron microscopy image of LDH/Ap_4A nanoparticles.

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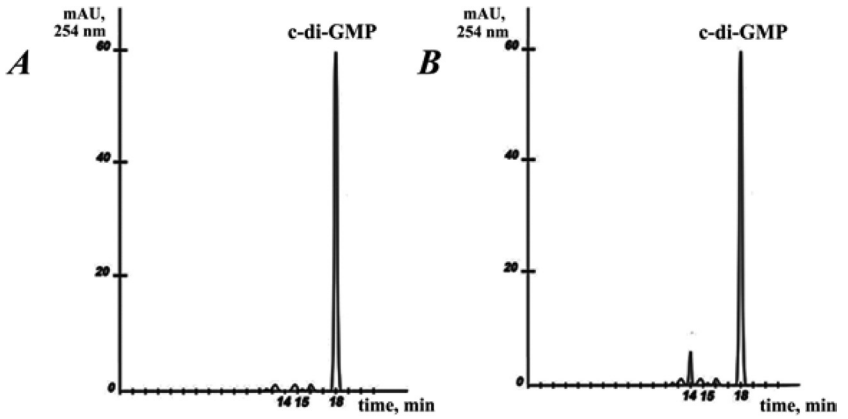


FIGURE 19.4 Purity of c-di-GMP tested by HPLC. (A) HPLC profile of c-di-GMP before intercalation into LDH. (B) HPLC profile of c-di-GMP after elution from LDH/dinucleotide nanocomposite.

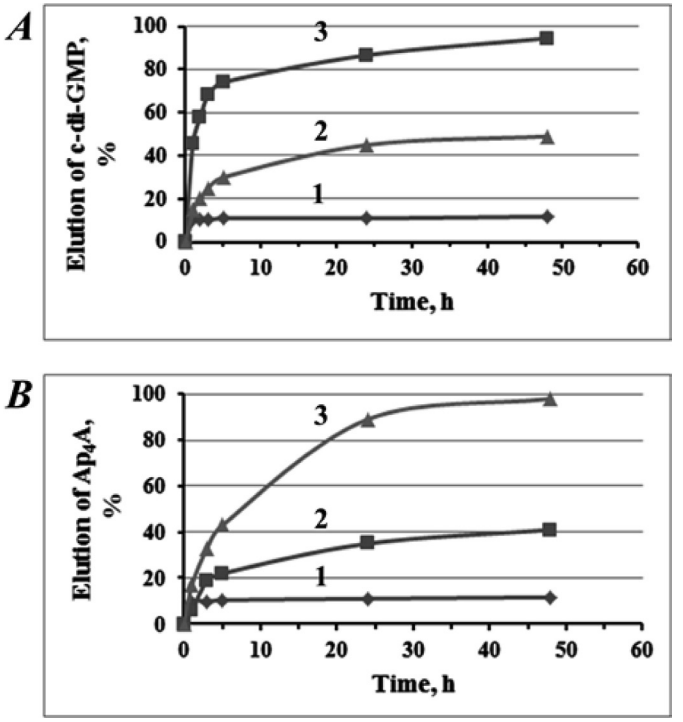


FIGURE 19.5 Release of c-di-GMP (A) and Ap_4A (B) from LDH/dinucleotide nanocomposites in H_2O (1) and 0.05 M phosphate-citrate buffer solution at pH 7.5 (2) and pH 4.4 (3).

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that the release of Ap_4A and c-di-GMP from the nanohybrids is evidently pH dependent, and its rate at pH 7.5 is remarkably lower than that at pH 4.4.

Such a discrepancy in the release rate at pH 4.4 and pH 7.5 may be due to a possible difference in mechanism responsible for the dissociation of the guest molecules from the nanohybrid [25]. At acidic pH, LDHs begin to dissolve. This would indicate that release of an interlayer molecule should occur mainly through the removal of inorganic host. At pH above 7.0, LDHs are likely to be more stable, and as a result, release may be attributed to the restricted motion of guest molecules arising from steric effect of LDHs and the electrostatic interaction between the anions and positively charged LDHs layers. That is to say, the mechanism of release in the pH 4.4 environment should engage both dissolution of LDH layers and ion exchange, while at pH 7.5 the mechanism should include mainly ion exchange of gallery and the buffer solution [25]. The lower detachment rate of Ap_4A and c-di-GMP from the nanohybrids at pH 7.5 indicates that LDH nanocomposites are indeed an excellent potential drug delivery system.

Summing up, this study proved the first successful demonstration that Ap_4A and c-di-GMP may be reversibly intercalated into LDH. Dinucleotide-LDH nanohybrid appears to be an attractive carrier for passive drug delivery to the target tissues.

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KEYWORDS

- **Al, Mg layered double hydroxides (LDHs)**
- **bis-(3′-5′)-cyclic dimeric deoxyguanosine monophosphate (c-di-GMP)**
- **diadenosine-5′,5‴-P1**
- **nanocomposites**
- **P4-tetraphosphate (Ap_4A)**

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